

Chapter 4: Network Layer

Chapter goals:

- ❑ understand principles behind network layer services:
 - network layer service models
 - forwarding versus routing
 - how a router works
 - addressing and routing (path selection)
 - dealing with scale
 - advanced topics: IPv6, multicast
- ❑ instantiation, implementation in the Internet

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Chapter 4: Network Layer

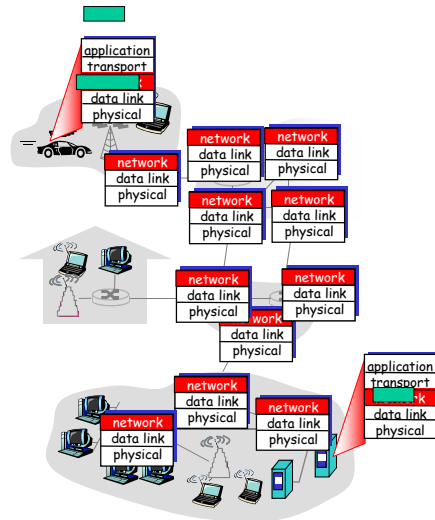
- ❑ 4.1 Introduction
- ❑ 4.2 Virtual circuit and datagram networks
- ❑ 4.3 What's inside a router
- ❑ 4.4 IP: Internet Protocol
 - Datagram format
 - IPv4 addressing
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Network layer

- ❑ delivers segments from sending to receiving host
 - sender encapsulates segments into datagrams
 - Receiver de-encapsulates and delivers segments to transport layer
- ❑ network layer in **every** host, **every** router
- ❑ router examines IP header field in every passing datagram (exception: routers running MPLS)



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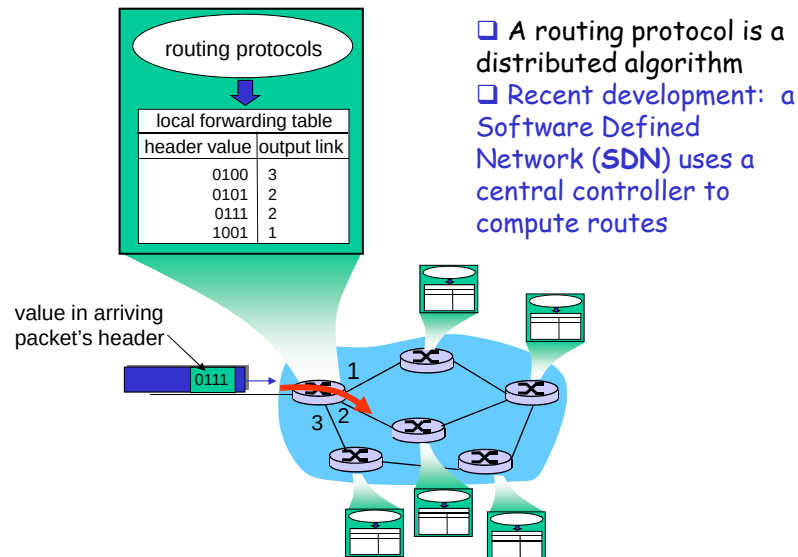
Key Network-Layer Functions

- ❑ **forwarding**: move a packet from router's input interface to an appropriate output interface
- ❑ **routing**: determine route taken by packets from source to destination
 - *global address space and a packet format*
 - *routing protocols (intra-network and inter-network)*

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Interplay between routing and forwarding



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Virtual circuit networks need 3rd function

- Before datagrams can flow, end hosts and routers between them establish a *virtual circuit*
 - Routers maintain state info
 - Following networks designed initially to compete with IP:
 - ATM, frame relay, X.25*
 - Designed more recently to enhance IP: *MPLS*
- Virtual circuits of these networks serve as **virtual links** in Internet

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Network layer service models

Q: What *service model* for delivering datagrams from sender to receiver?

Desired services for individual datagrams (connectionless):

- ❑ reliable delivery
- ❑ guaranteed delivery in less than some bound, for example, 40 msec delay

Desired services for a flow of datagrams (connection-oriented):

- ❑ reliable, in-order datagram delivery
- ❑ guaranteed minimum bandwidth to flow
- ❑ bound on fluctuations in inter-packet spacing (jitter)

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Network layer service models:

Network Architecture	Service Model	Guarantees ?				Congestion feedback
		Bandwidth	Loss	Order	Timing	
Internet	best effort	none	no	no	no	no (inferred via loss)
ATM	CBR	constant rate	yes	yes	yes	no congestion
ATM	VBR	guaranteed rate	yes	yes	yes	no congestion
ATM	ABR	guaranteed minimum	no	yes	no	yes
ATM	UBR	none	no	yes	no	no

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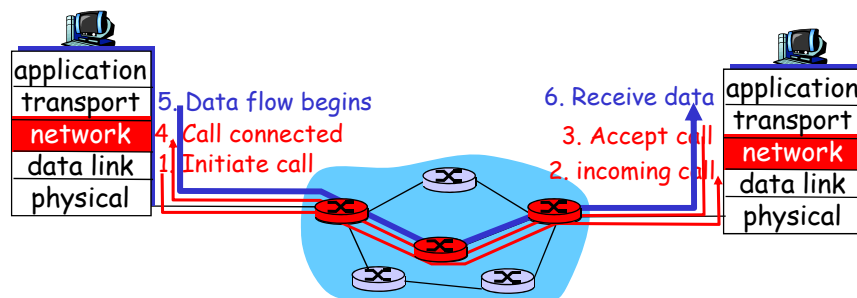
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Virtual circuits: signaling protocols

- ❑ used to set up, maintain, tear down VC
- ❑ not used in today's Internet (but may be used underneath the IP layer to provide a virtual link)



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Virtual circuit (VC)

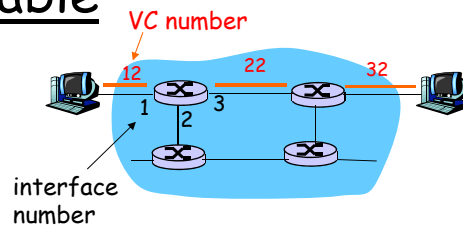
- ❑ call setup, teardown for each call *before* data can flow
- ❑ each packet carries a VC identifier which
 - is fixed length and short
 - only needs to be unique for a link
- ❑ *every* router on source-dest path **maintains state information** for each passing VC
 - incoming and outgoing VC identifiers,
 - resources allocated to VC (bandwidth, buffers)

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VC Forwarding table

Forwarding table in
northwest router:



Incoming interface	Incoming VC #	Outgoing interface	Outgoing VC #
1	12	3	22
2	63	1	18
3	7	2	17
1	97	3	87
...	...	...	...

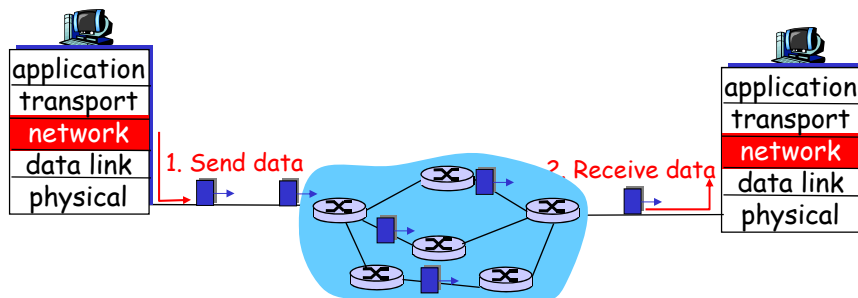
Forwarding is fast because short fixed-length VC numbers are used vs. IP forwarding table with variable-length prefixes

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Datagram networks

- ❑ no network-level concept of "connection"
- ❑ each packet forwarded independently using destination host address
 - packets between same source-dest pair may take different paths



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Forwarding table

4 billion
possible entries

<u>Destination Address Range</u>	<u>Link Interface</u>
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	2
otherwise	3

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Longest prefix match

<u>Prefix Match</u>	<u>Link Interface</u>
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
otherwise	3

Examples

DA: 11001000 00010111 00010110 10100001 Which interface?

DA: 11001000 00010111 00011000 10101010 Which interface?

A forwarding table in an Internet core router has about 300,000 IP prefixes

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Origins of datagram and VC

Internet (datagram)

- data exchange between computers
 - "elastic" service, no strict timing requirement
- many link types
 - different characteristics
 - uniform service difficult
- "smart" end systems (computers)
 - can adapt, perform control, error recovery
 - *simplicity* inside network, *complexity* at "edge"

ATM (VC)

- evolved from telephony
- human conversation:
 - strict timing, reliability requirements
 - need for guaranteed services
- "dumb" end systems
 - telephones
 - complexity inside network

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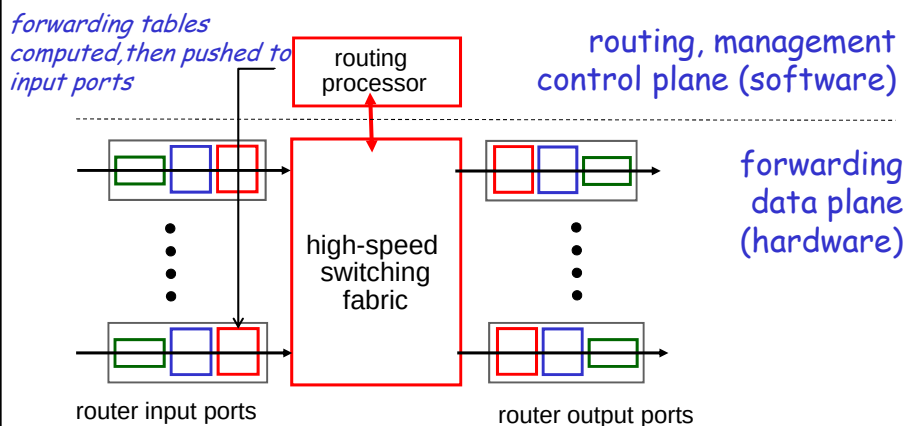
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Router architecture overview

Two key router functions:

- ❖ run routing protocols (RIP, OSPF, BGP)
- ❖ *forward* datagrams from incoming to outgoing link



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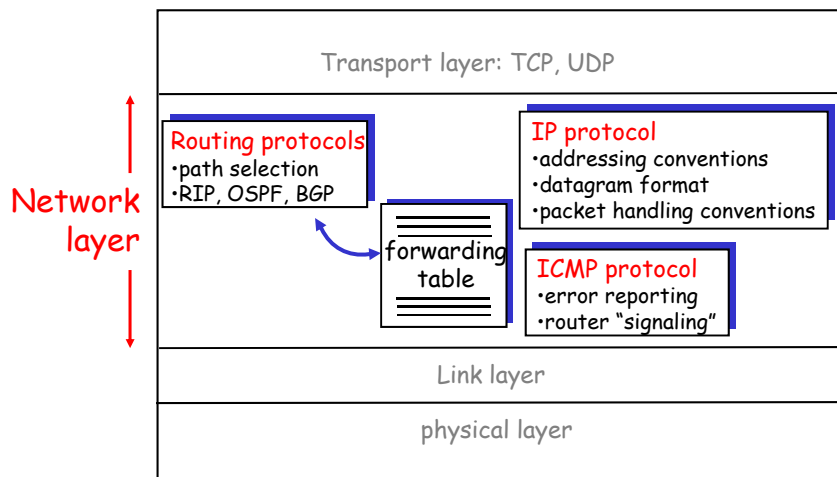
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The Internet Network layer

Host, router network layer functions:



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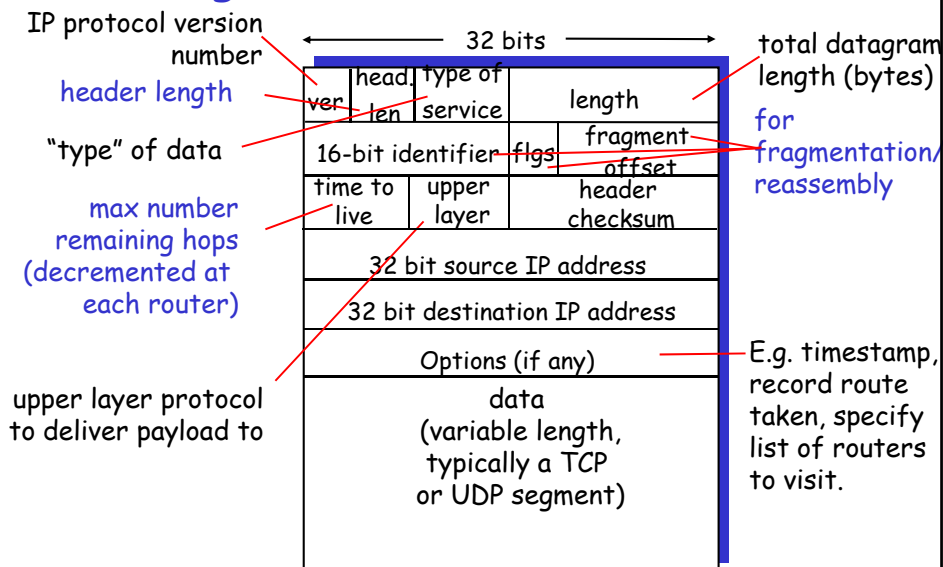
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IP datagram format

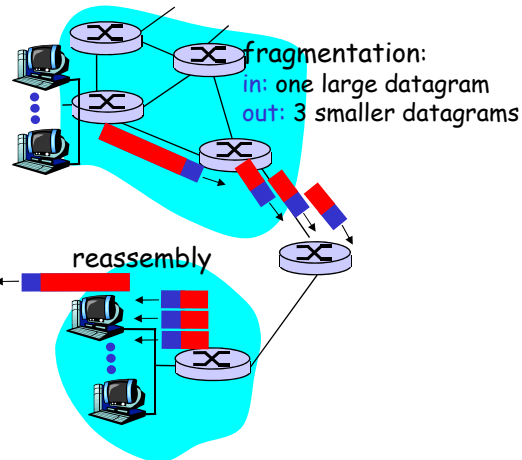


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IP Fragmentation & Reassembly

- MTU (max.transfer size)
 - different link types, different MTUs
 - Support MTU of at least 576 bytes
- too large IP datagram "fragmented" within net
 - reassembled only at final destination
 - IP header bits used to identify, order related fragments



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IP Fragmentation and Reassembly

3980 bytes of data

Example

- 4000 byte datagram
- MTU = 1500 bytes

1480 bytes in data field

offset =
 $1480/8$

length	ID	fragflag	offset
=4000	=x	=0	=0

One large datagram becomes several smaller datagrams

length	ID	fragflag	offset
=1500	=x	=1	=0
=1500	=x	=1	=185
=1040	=x	=0	=370

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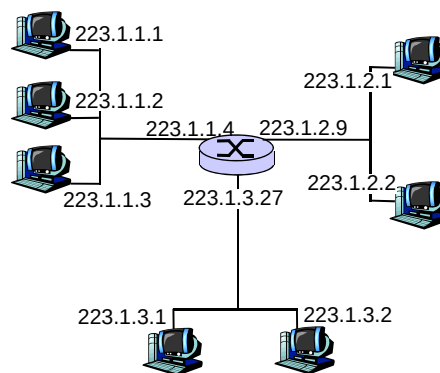
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IP Addressing: introduction

- ❑ **IP address:** 32-bit identifier for an *interface*
- ❑ **interface:** connection between host/router and physical link (*wired or wireless*)
 - a router typically has *multiple* interfaces
 - a host typically has *one* interface



Dotted decimal notation

223.1.1.1 = 11011111 00000001 00000001 00000001
 223 1 1 1

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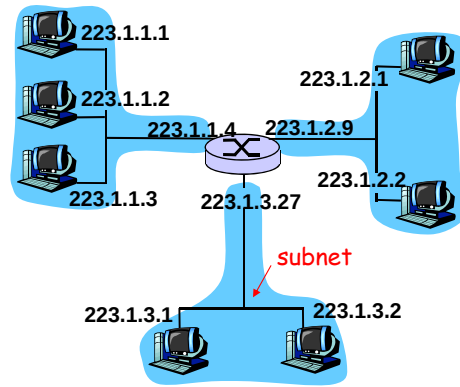
Subnets

□ IP address:

- **subnet** part (high order bits)
- **host** part (low order bits)

□ What's a subnet ?

- device interfaces with same subnet part of IP address
- **can physically reach each other without a router**



network consisting of 3 subnets

Notes: (1) Layer-2 devices are part of link; (2) Virtual LAN (VLAN) - see Chapter 5

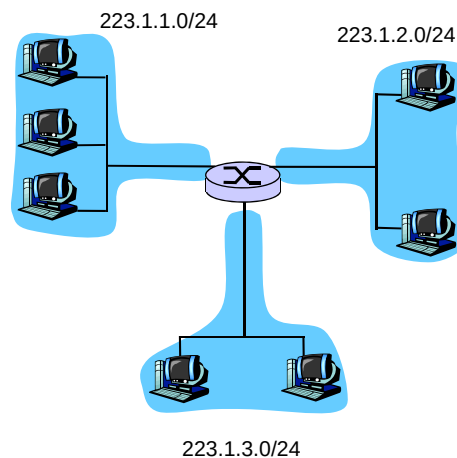
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Subnets

Recipe

- To determine the subnets, detach each interface from its host or router, creating islands of isolated networks. Each isolated network is a **subnet**.



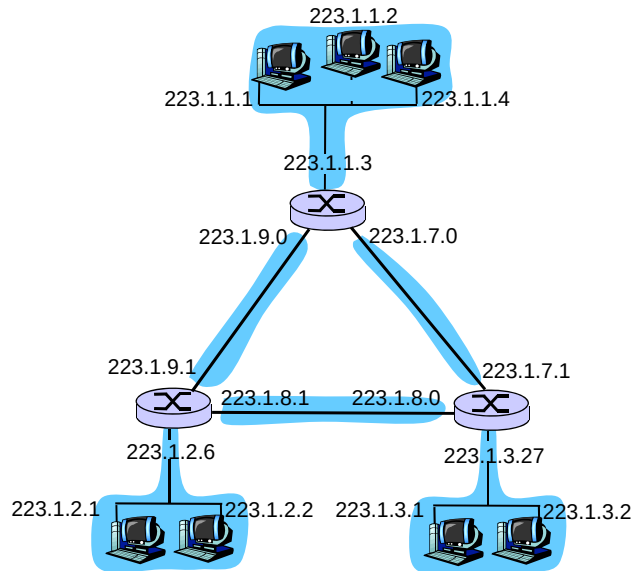
Subnet mask: /24

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Subnets

How many?



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IP addressing: CIDR

CIDR: Classless InterDomain Routing

- subnet portion of address of **variable length**
- address format: **a.b.c.d/x**, where x is # bits in subnet portion of address

← subnet part → host part
11001000 00010111 00010000 00000000

200.23.16.0/23

Classful addressing (now obsolete): fixed-length subnet portion of 8, 16 or 24 bits

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IP addresses: how to get one?

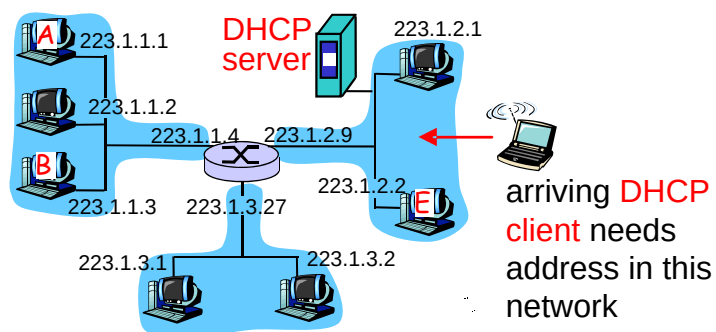
Q: How does *host* get IP address?

- hard-coded by system admin in a file
- **DHCP**: Dynamic Host Configuration Protocol:
dynamically get address from a server
 - "plug-and-play"

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DHCP client-server scenario



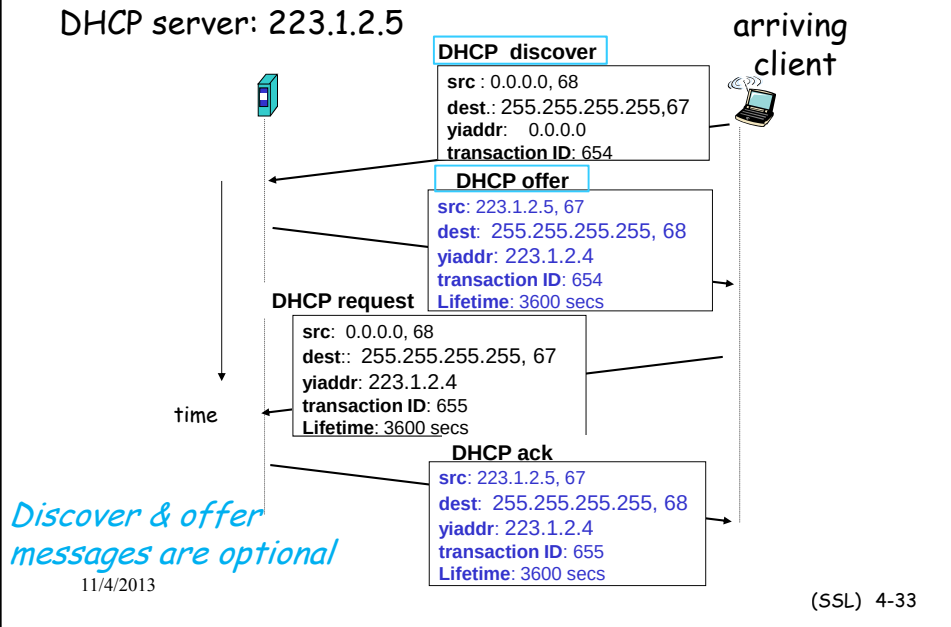
A router may act as a relay agent

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DHCP client-server scenario

DHCP server: 223.1.2.5



DHCP: more than IP address

DHCP can return more than just an allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating subnet portion of address)

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IP addresses: how to get them?

- ❑ **ICANN** (Internet Corporation for Assigned Names and Numbers)/IANA (Internet Assigned Numbers Authority)
 - **allocates IP addresses** (IPv4 address exhaustion on 1/31/2011)
 - **oversees DNS**
 - **assigns domain names, resolves disputes**
- ❑ **Regional, national, and local Internet registries, and ISPs**
 - **End-user organization can be assigned IP address space from one of the above**

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IP address prefix: how to get one?

A: Typically, a customer network gets allocated a portion of its provider ISP's address space

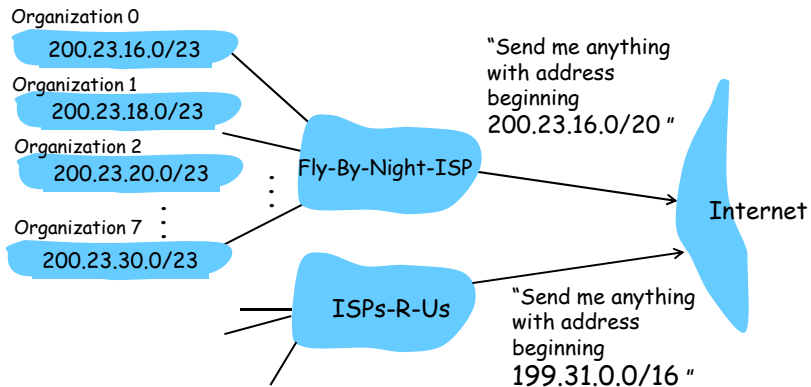
ISP's block	<u>11001000</u>	<u>00010111</u>	<u>00010000</u>	00000000	200.23.16.0/20
Organization 0	<u>11001000</u>	<u>00010111</u>	<u>00010000</u>	00000000	200.23.16.0/23
Organization 1	<u>11001000</u>	<u>00010111</u>	<u>00010010</u>	00000000	200.23.18.0/23
Organization 2	<u>11001000</u>	<u>00010111</u>	<u>00010100</u>	00000000	200.23.20.0/23
...				...	
Organization 7	<u>11001000</u>	<u>00010111</u>	<u>00011110</u>	00000000	200.23.30.0/23

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Hierarchical addressing: route aggregation

allows efficient advertisement of routing information

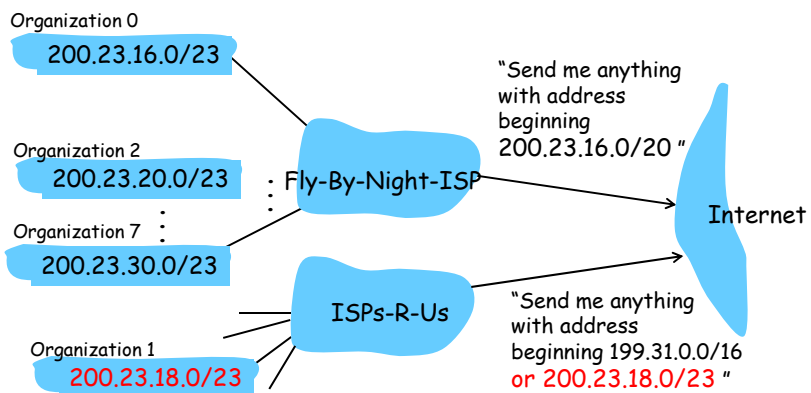


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Hierarchical addressing: more specific routes

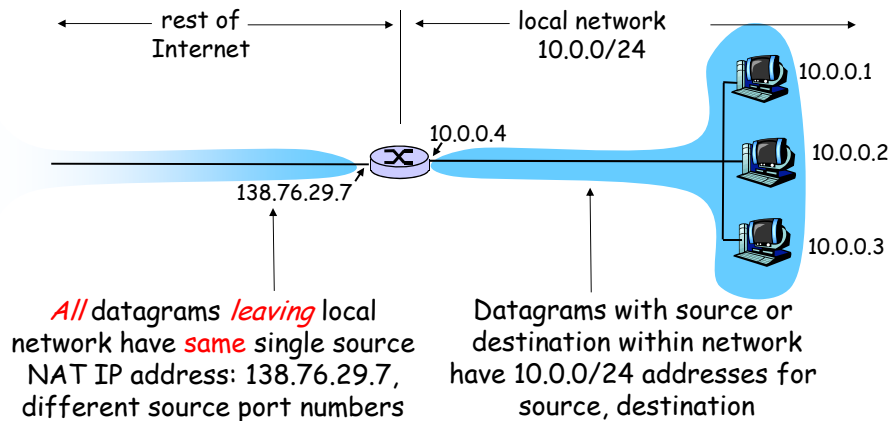
ISPs-R-Us has a more specific route to Organization 1
this is the reason for longest prefix match



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NAT: Network Address Translation



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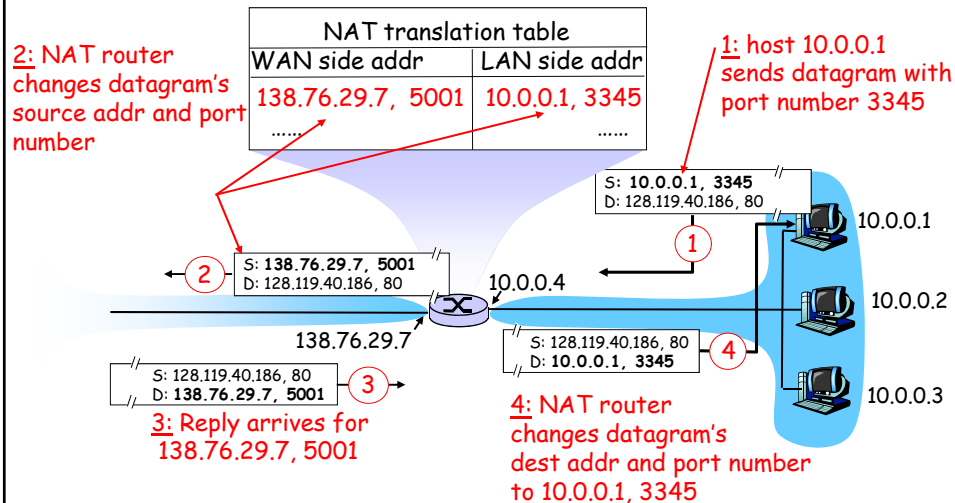
NAT: Network Address Translation

- **Motivation:** local network uses just one IP address as far as outside world is concerned
 - can change addresses of devices in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - devices inside local net not explicitly addressable/visible by outside world (a security plus).

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NAT: Network Address Translation



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NAT: Network Address Translation

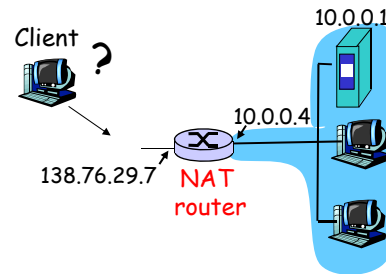
- ❑ 16-bit port-number field:
 - 60,000 simultaneous connections with a single IP address
- ❑ NAT is controversial:
 - routers should only process up to layer 3
 - violates "end-to-end argument"
 - NAT possibility must be taken into account by app designers, eg, IPsec, P2P applications
 - address shortage should instead be solved by IPv6

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NAT traversal problem

- ❑ client wants to connect to server with address 10.0.0.1
 - only one externally visible IP address: 138.76.29.7
- ❑ *statically* configure NAT to forward incoming connection requests at given port to server
 - e.g., (123.76.29.7, port 2500) always forwarded to 10.0.0.1 port 2500
- ❑ use UPnP IGD protocol to configure
 - Vulnerable to access/attack from the Internet

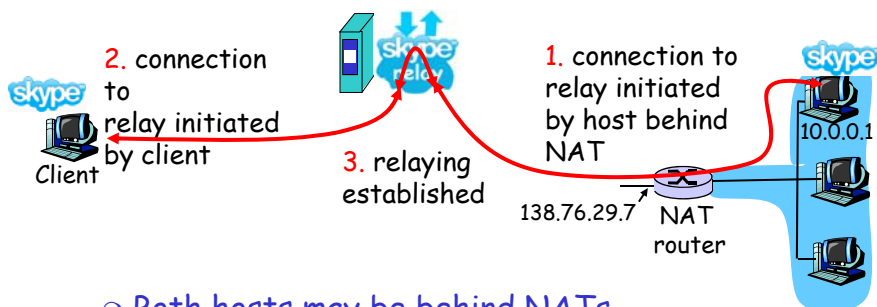


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NAT traversal problem

- ❑ solution 3: relaying (used in Skype)
 - relay bridges packets between to connections



- Both hosts may be behind NATs

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ICMP: Internet Control Message Protocol

- ❑ "above" IP in network layer
 - ICMP msgs carried in IP datagrams
 - error reporting: unreachable host, network, port, protocol
 - echo request/reply (used by ping)
 - ❑ ICMP message
 - type, code plus first 8 bytes of IP datagram causing error
- | Type | Code | description |
|------|------|---|
| 0 | 0 | echo reply (ping) |
| 3 | 0 | dest. network unreachable |
| 3 | 1 | dest host unreachable |
| 3 | 2 | dest protocol unreachable |
| 3 | 3 | dest port unreachable |
| 3 | 6 | dest network unknown |
| 3 | 7 | dest host unknown |
| 4 | 0 | source quench (congestion control - not used) |
| 8 | 0 | echo request (ping) |
| 9 | 0 | route advertisement |
| 10 | 0 | router discovery |
| 11 | 0 | TTL expired |
| 12 | 0 | bad IP header |

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Traceroute uses ICMP messages

- ❑ Source sends series of UDP segments to dest
 - First has TTL =1
 - Second has TTL=2, ..., each with unlikely port number
- ❑ When nth datagram arrives to nth router:
 - Router discards datagram and
 - sends to source a "TTL expired" message with name of router & IP address
- ❑ When "TTL expired" message arrives, source calculates RTT
- ❑ Traceroute does this 3 times for each TTL value
- Stopping criterion
- ❑ Such a UDP segment arrives at destination host
- ❑ Destination returns msg "dest port unreachable" packet
- ❑ Upon receipt of this msg, source stops.

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IPv6

- ❑ **Initial motivation:** 32-bit address space soon to be completely allocated.
- ❑ **Additional motivation:**
 - simpler header format to speed up processing/forwarding
 - header change to facilitate QoS
- ❑ **IPv6 datagram format:**
 - fixed-length 40 byte header
 - no fragmentation allowed

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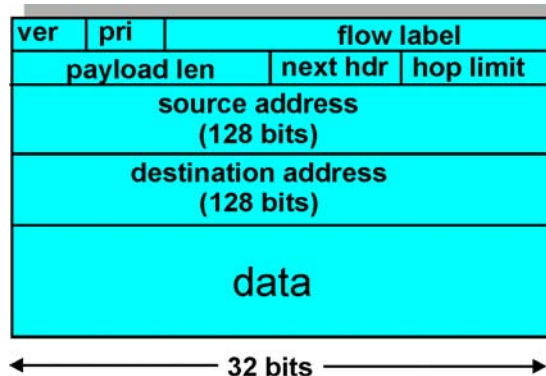
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IPv6 Header (Cont)

Priority: identify priority among datagrams in flow

Flow Label: identify datagrams in same "flow."
(concept of "flow" not defined).

Next header: identify upper layer protocol for data



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Other Changes from IPv4

- ❑ *Checksum*: removed entirely to reduce processing time at each hop
- ❑ *Options*: allowed, but outside of header, indicated by "Next Header" field
- ❑ *ICMPv6*: new version of ICMP
 - additional message types, e.g. "Packet Too Big"
 - including *multicast group management* functions

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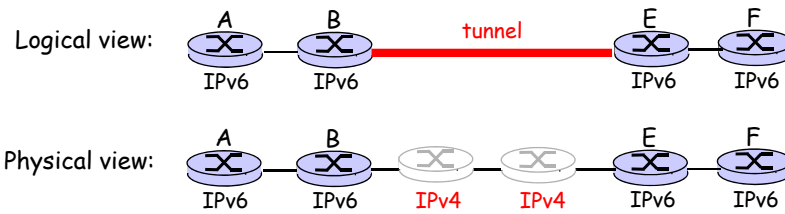
Transition From IPv4 To IPv6

- ❑ Not all routers can be upgraded simultaneous
 - no "flag day"
 - How will the network operate with mixed IPv4 and IPv6 routers?
- ❑ *Tunneling*: IPv6 carried as payload in IPv4 datagram among IPv4 routers

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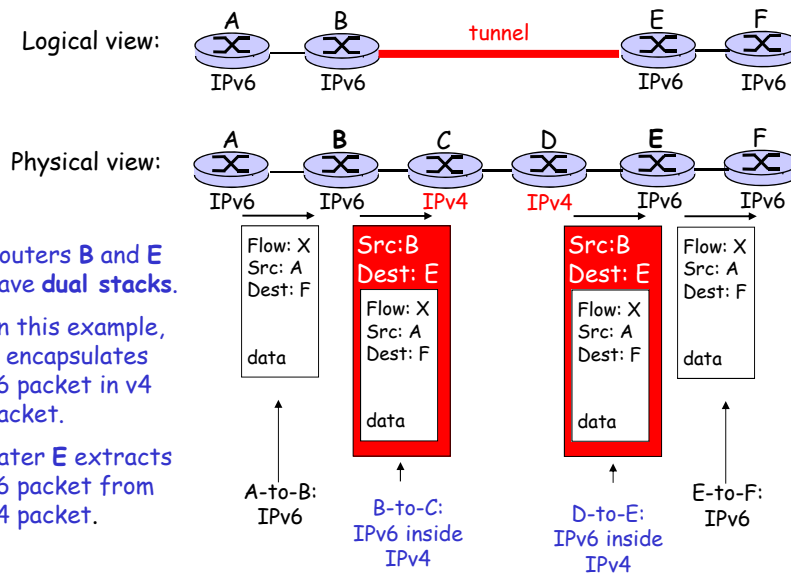
Tunneling



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Tunneling



Routers B and E have dual stacks.

In this example, B encapsulates v6 packet in v4 packet.

Later E extracts v6 packet from v4 packet.

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Concept - Tunnel as a virtual link

Many possibilities:

- ❑ IPv6 in IPv4 tunnel (previous example)
- ❑ IPv4 in IPv6 tunnel
- ❑ IPv4 in IPv4 tunnel
- ❑ IPv4 in MPLS tunnel

...

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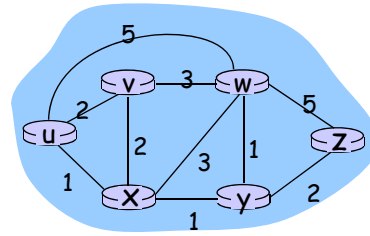
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Graph abstraction

Graph: $G = (N, E)$

N = set of routers = $\{ u, v, w, x, y, z \}$

E = set of links = $\{ (u,v), (u,x), (v,x), (v,w), (x,w), (x,y), (w,y), (w,z), (y,z) \}$



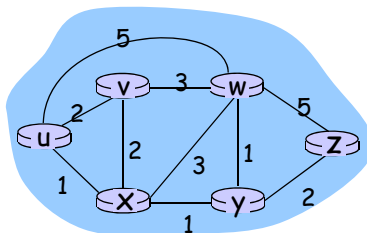
Remark: Graph abstraction is also useful in other network contexts

Example: P2P, where N is set of peers and E is set of TCP connections

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Network Layer (SSL) 4-57

Graph abstraction: link costs



o $c(x, x')$ = cost of link (x, x')

o cost could be 1, or inversely related to bandwidth, or inversely related to "congestion", etc.

Cost of path $(x_1, x_2, x_3, \dots, x_p) = c(x_1, x_2) + c(x_2, x_3) + \dots + c(x_{p-1}, x_p)$

Routing protocol tries to find least-cost path

Comments:

- full-duplex links may not have symmetric costs
- cost of path computation is *ad hoc* if the link cost metric is not additive

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Routing Algorithm classification

Global or decentralized information?

Global info:

- ❑ all routers have complete topology, link costs
- ❑ **link state** protocols

Decentralized info:

- ❑ router knows physically-connected neighbors, link costs to neighbors
- ❑ **distance vector** protocols

Static or dynamic?

- ❑ **Static** - only after topology change

- ❑ **Dynamic**

- periodic update
- in response to link cost changes
- may result in route flaps

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Network Layer (SSL) 4-59

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A Link-State Routing protocol

- net topology, link costs known to every node
 - accomplished via **link state broadcast**
 - all nodes have same info

Dijkstra's algorithm

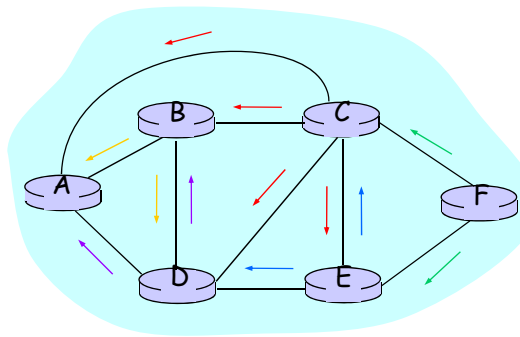
- computes least cost paths from one node ("source") to all other nodes in a graph
 - *iterative*: after k iterations, source knows least-cost paths to k destinations
 - yields **forwarding table** for source node

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Network Layer (SSL) 4-61

Link State Broadcast

- Flooding
 - Source node of "link state" sends packet to all neighbors
 - Intermediate node resends to neighbors except where packet arrived
 - Many duplicates which must be recognized by nodes



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Distance Vector Algorithm basis

Bellman-Ford Equation (dynamic programming)

Define

$d_x(y) :=$ cost of least-cost path from x to y

Then

$$d_x(y) = \min_v \{c(x,v) + d_v(y)\}$$

where min is taken over all neighbors v of x

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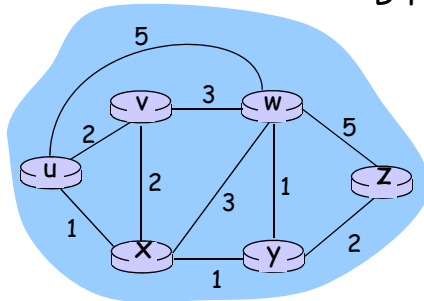
Network Layer (SSL) 4-64

Bellman-Ford example

Clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

$$d_u(z) = \min \{ c(u,v) + d_v(z), \\ c(u,x) + d_x(z), \\ c(u,w) + d_w(z) \} \\ = \min \{ 2 + 5, \\ 1 + 3, \\ 5 + 3 \} = 4$$



The node that achieves minimum is next hop in shortest path → put it in forwarding table

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Distance Vectors Protocol (1)

□ Node x

- knows cost to each neighbor v: $c(x,v)$
- sends its own distance vector (DV) estimate $[D_x(y): y \in N]$ to its neighbors periodically where $D_x(y)$ denotes **estimate** of least cost from x to y

□ From each neighbor v, x receives

$[D_v(y): y \in N]$

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Distance Vector Protocol (2)

- When a node x receives a **new DV estimate** from a neighbor, it updates its own DV estimate using B-F equation:

$$D_x(y) \leftarrow \min_v \{c(x,v) + D_v(y)\} \quad \text{for each node } y \in N$$

If the v that achieves least cost to y is new, node x updates its forwarding table and DV

- Eventually, assuming that link costs and topology do not change, the estimate $D_x(y)$ converges to the actual least cost $d_x(y)$ for all x, y

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Distance Vector Protocol - summary

**Distributed,
asynchronous,
iterative**

Initially, $D_x(y) = c(x,y)$ if x and y are direct neighbors; otherwise, $D_x(y) = \infty$

Each node:

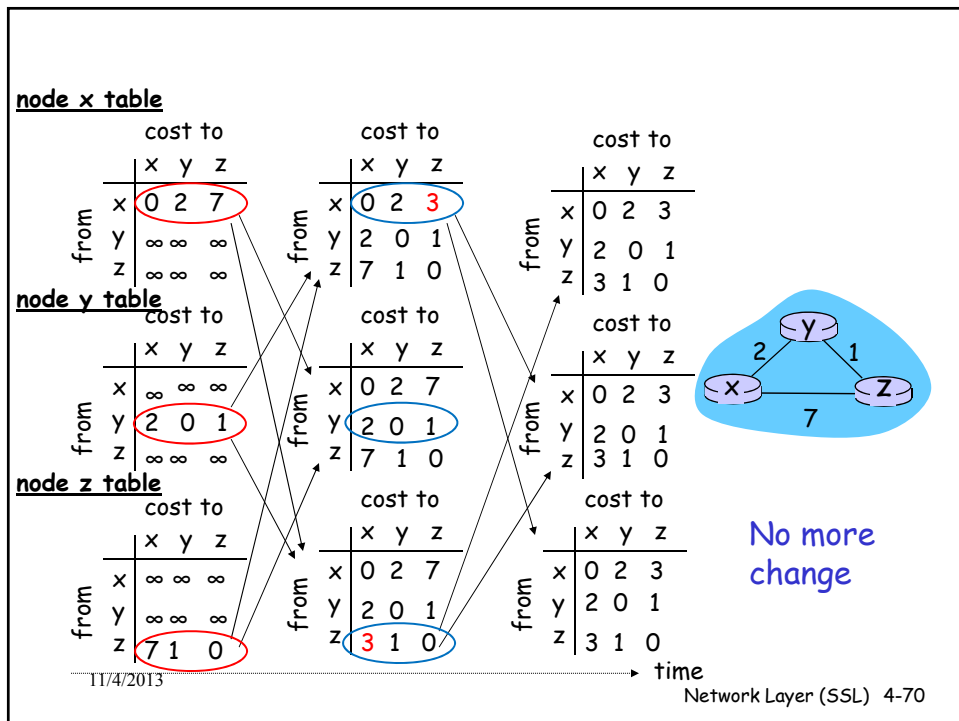
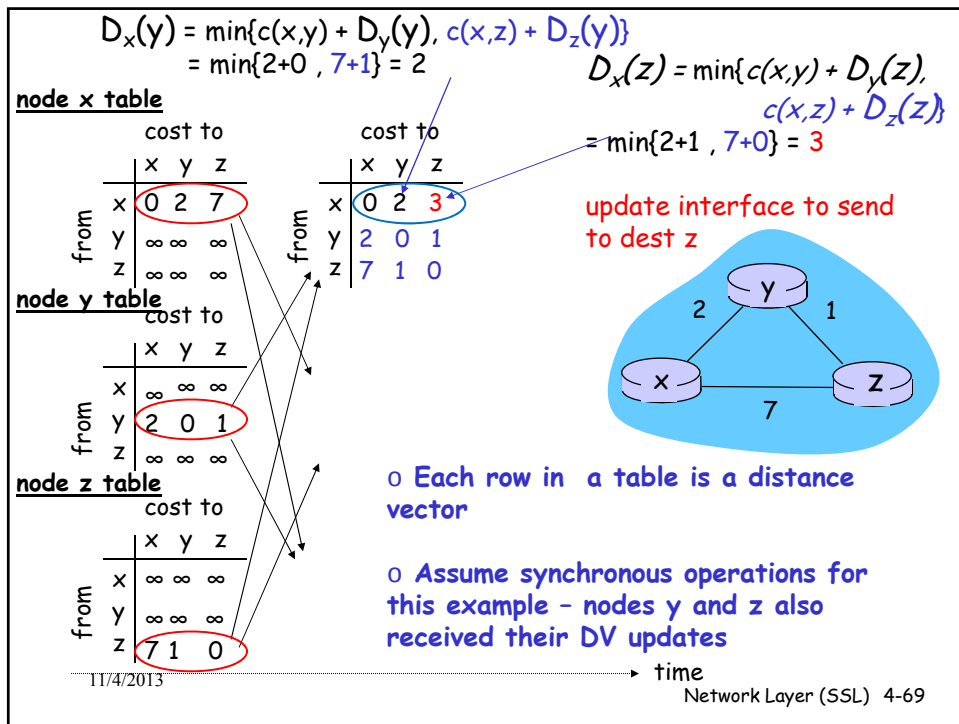
waits for a change in local link cost or a msg from a neighbor

recomputes estimates

if DV estimate for any dest has changed, updates its own state and **notifies** its neighbors

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Distance Vector: good news travels fast

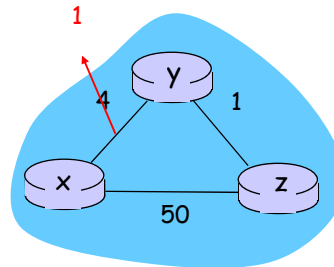
y detects a lower link cost to x, updates its DV, and sends new DV to node **z**.

z receives **y**'s updated DV, updates its own DV, and sends new DV to its neighbors.

later, **y** receives **z**'s updated DV. **y**'s least cost does not change.

A similar interaction between nodes **x** and **z**.

The DV protocol converges quickly.



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Distance Vector: "count to infinity" problem

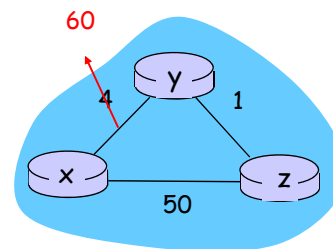
Link cost increase:

- ❑ Y still has stale information saying that it can go to X via Z in 6
- ❑ 44 iterations (msg exchanges between y and z) before protocol stabilizes

Poisoned reverse:

- ❑ If Z routes through Y to get to X:
 - Z tells Y its (Z's) distance to X is infinite (so Y won't route to X via Z)
- ❑ will this completely solve count to infinity problem?

bad news travels slowly!



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Hierarchical network

Our routing study thus far

- ❑ all routers identical
- ❑ network "flat"

... *not* true in practice

scale: hundreds of millions destinations:

- ❑ can't store all dest's in forwarding tables
- ❑ Link State and Distance Vector do not scale

administrative autonomy

- ❑ internet is a network of networks
- ❑ each network admin may want to control routing in its own network

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Hierarchical Routing

- aggregate routers into regions, **autonomous systems (AS)**
 - Gateway router
 - has direct link to a router in another AS
- routers in same AS run same routing protocol
 - **intra-AS routing protocol**
 - routers in different ASes can run different intra-AS routing protocols

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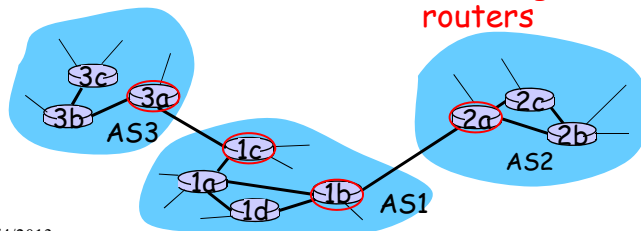
Inter-AS routing

- suppose router in AS1 receives datagram destined outside of AS1:
 - router should forward packet to a neighbor AS, but which one?

AS1 must:

1. learn which dests are reachable through AS2, which through AS3
2. propagate this reachability info to all routers in AS1

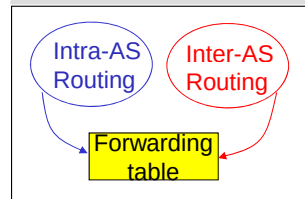
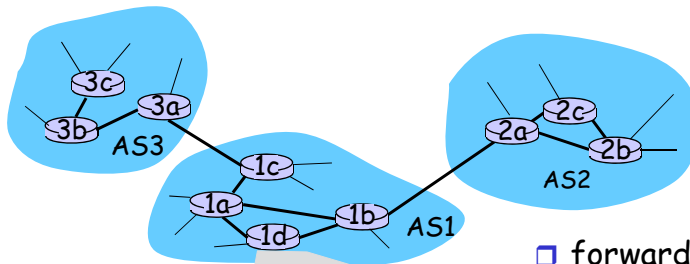
Job of inter-AS routing is performed by **border gateway routers**



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Interconnected ASes



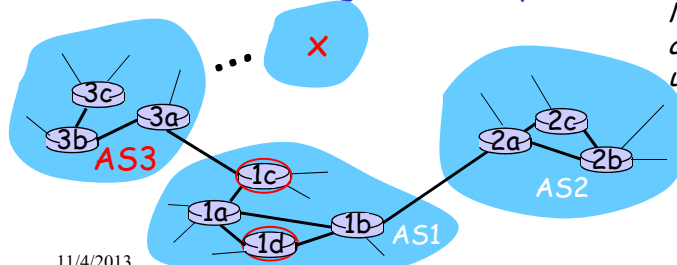
- forwarding table configured by both intra- and inter-AS routing protocols
 - intra-AS protocol sets entries for internal dests
 - inter-AS & intra-AS protocols set entries for external dests

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Example: Setting forwarding table in router 1d

- AS1 learns from inter-AS protocol that subnet **x** is reachable via **AS3** (gateway **1c**) but not via AS2.
- **inter-AS** protocol propagates reachability info to all internal routers.
- for subnet **x**, router **1d** determines from **intra-AS** routing info that its interface **I** is on the least cost path to **1c**
 - installs forwarding table entry (**x, I**)



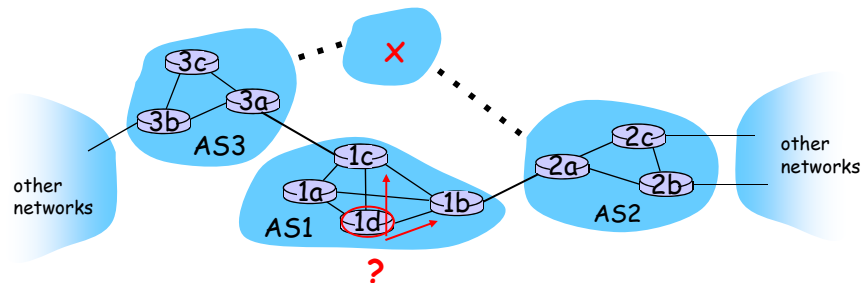
Note: both inter-AS and intra-AS are used

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Example: Choosing among multiple ASes

- now suppose AS1 learns from inter-AS protocol that subnet **x** is reachable from AS3 *and* from AS2.
- to configure forwarding table, router **1d** needs to know which gateway it should forward packets for dest **x**
 - this is also a task of inter-AS routing protocol



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Intra-AS Routing

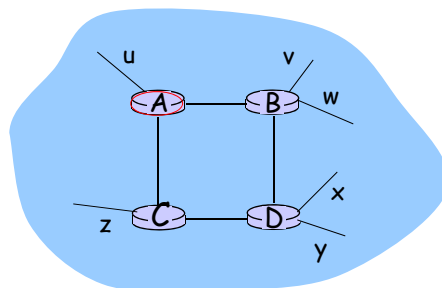
- ❑ also known as **Interior Gateway Protocols (IGP)**
- ❑ most common Intra-AS routing protocols:
 - **RIP**: Routing Information Protocol
 - **OSPF**: Open Shortest Path First
 - **EIGRP** (Cisco) - distance vector with "loop-freedom"

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RIP (Routing Information Protocol)

- ❑ **distance vector** algorithm
- ❑ included in BSD-UNIX Distribution in 1982
- ❑ distance metric: # of hops (max = 15 hops)



From router A to subnets:

<u>destination</u>	<u>hops</u>
u	1
v	2
w	2
x	3
y	3
z	2

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RIP advertisements

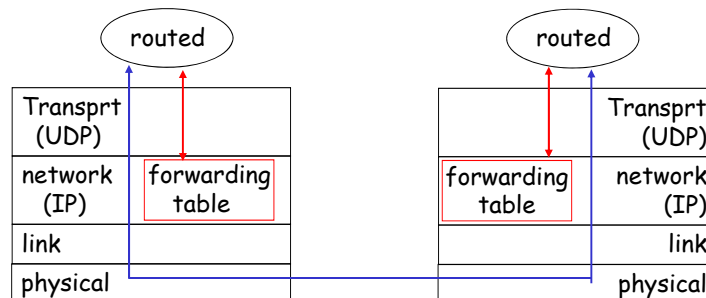
- ❑ distance vectors: exchanged with neighbors every 30 sec via Response Message (also called **advertisement**)
- ❑ each advertisement: list of up to 25 destination subnets within AS

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RIP Table processing

- ❑ RIP routing tables managed by **application-level** process called **routed** (daemon)
- ❑ advertisements sent in **UDP packets**, periodically repeated



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OSPF (Open Shortest Path First)

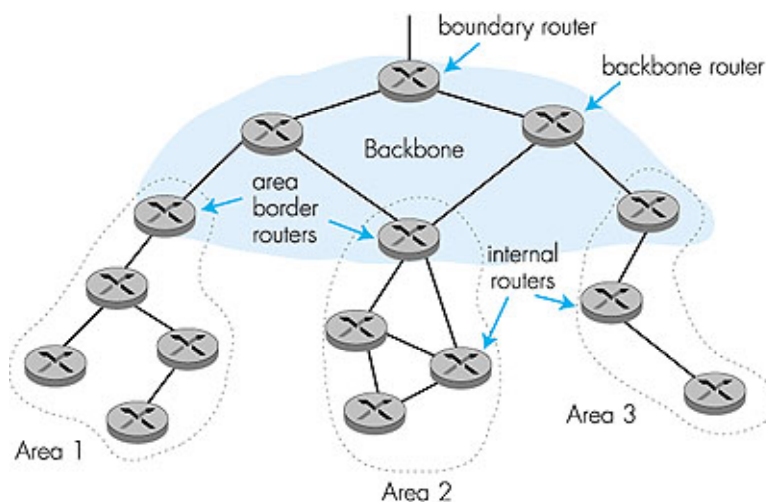
- ❑ "open": publicly available
- ❑ uses **Link State** algorithm
- ❑ OSPF advertisement carries one entry per neighbor router
- ❑ advertisements disseminated to **entire** AS (via flooding)
 - carried in OSPF messages **directly over IP** (rather than TCP or UDP)
- ❑ **security**: all OSPF messages authenticated
- ❑ ...

Note: IS-IS routing protocol: nearly identical to OSPF

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Hierarchical OSPF



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Internet inter-AS routing: BGP

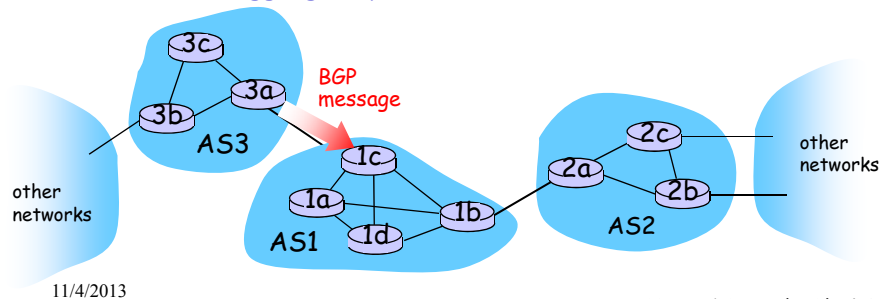
- ❑ BGP (Border Gateway Protocol): *the de facto standard*
- ❑ an AS advertises its existence to rest of Internet: "*I am here*" and
 1. obtains reachability information from neighboring ASes (using eBGP)
 2. propagates reachability information to all internal routers of the AS (using iBGP)
 3. determines "good" routes to other ASes based on reachability information and policy.

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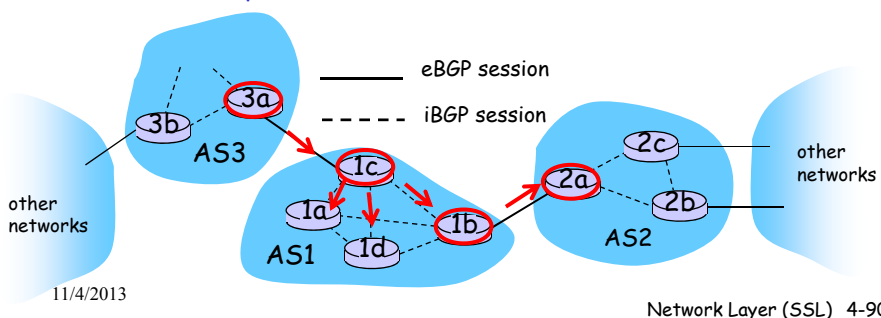
BGP basics

- **eBGP session:** two BGP routers ("peers") exchange messages over semi-permanent TCP connections
 - advertising *paths* to various destination network prefixes ("path vector" protocol)
- when AS3 advertises a prefix to AS1, AS3 *promises* it will forward datagrams towards that prefix
 - AS3 can aggregate prefixes in its advertisement



BGP basics: distributing path information

- using eBGP session between 3a and 1c, AS3 sends prefix reachability info to AS1.
 - 1c can then use **iBGP** to distribute new prefix info to all routers in AS1
- when a router learns of a new prefix, it creates entry for prefix in its forwarding table.
 - In this example, router 1b can then re-advertise new reachability info to AS2 over 1b-to-2a eBGP session



Path attributes & BGP routes

- ❑ advertised prefix includes BGP attributes.
 - prefix + attributes = "route"
- ❑ two important attributes:
 - **AS-PATH**: contains ASes through which prefix advertisement has passed: e.g, AS 67, AS 17
 - **NEXT-HOP**: the router interface (IP address) that begins the AS path
 - there may be multiple links from current AS to next-hop-AS
- ❑ when a gateway router receives route advertisement, it uses the AS's **import policy** to accept or decline.

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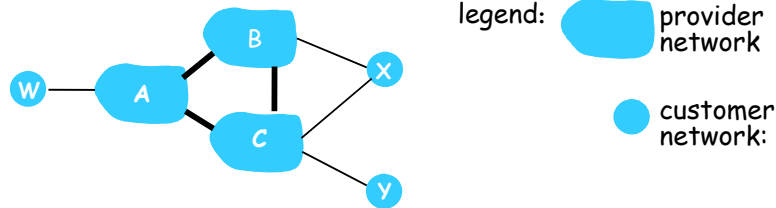
BGP route selection

- ❑ router may learn more than 1 route to some prefix. Router must select *one route*
- ❑ elimination rules:
 1. local preference value attribute (policy decision)
 2. shortest AS-PATH
 3. closest NEXT-HOP router: hot potato routing
 4. (additional criteria) ...

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BGP routing policy example

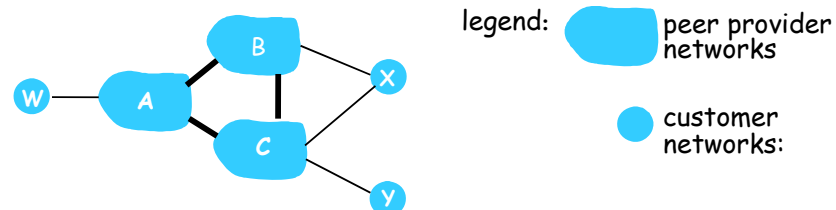


- A,B,C are **provider networks**
- X,W,Y are customers (of provider networks)
- X is **dual-homed**: attached to two networks
 - X does not want to route from B via itself to C
 - .. so X will not advertise to B that it has a route to C

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BGP routing policy example (2)



- A advertises path AW to B
- B advertises path BAW to X
- Should B advertise path BAW to C ?
 - No - B gets no "revenue" for routing CBAW since neither C nor A nor W is a customer of B
 - B wants to route **only to/from its customers**

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Why different Intra- and Inter-AS routing ?

Scale:

- ❑ hierarchical routing reduces table size, also update traffic

Policy:

- ❑ Inter-AS: admin wants control over how its traffic is routed, who routes through its net.
- ❑ Intra-AS: single admin, so no policy decisions needed

Performance:

- ❑ Intra-AS: can focus on performance
- ❑ Inter-AS: policy dominates performance

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Broadcast routing

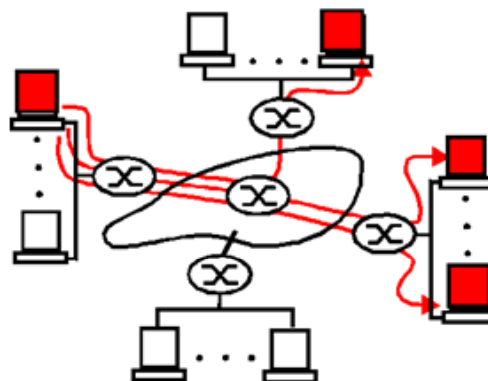
- ❑ **flooding**: each node sends a broadcast packet to all other neighbors
 - routing loops & broadcast storm - **not a solution**
- ❑ **controlled flooding** (some duplicates)
 - node sends to all other neighbors only if the packet is new
 - node keeps track of packet IDs already sent
 - used in Link State protocol
- ❑ **construct a spanning tree** first
 - no redundant packet received by any node
 - both communication and storage overheads

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Multicast - 3 approaches

1. **multiple unicasts**
 - source *duplication* as well as in-network *duplication*



multicast via unicast

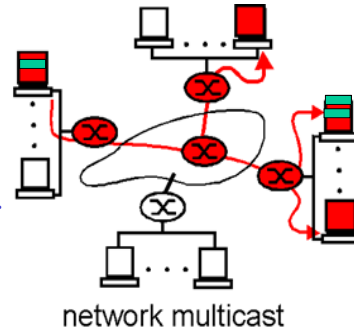
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Multicast - 3 approaches

2. network-supported multicast

- ❑ address indirection - multicast group address
- ❑ group management protocol - identify routers with multicast recipients
- ❑ *no duplication*



3. application-layer multicast

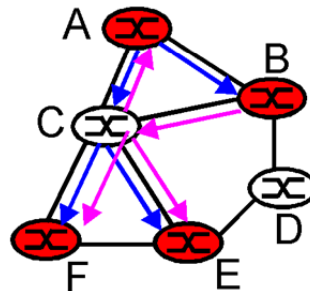
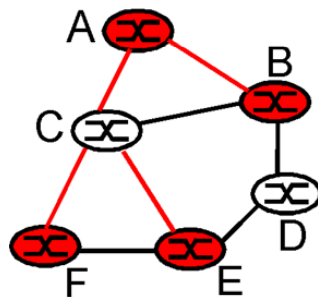
- ❖ *no source duplication*
- ❖ *in-network duplication*

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Multicast Tree options

- ❑ **GROUP SHARED TREE** (left figure) : single tree for all senders; bidirectional edges



- ❑ **SOURCE BASED TREE** (right figure): each source is the root of its own tree connecting to all members
 - *separate trees for different senders*

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End of Chapter 4

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